

EXP2-4 Blamestorming



The **aim** of EXP2-4 is to consider how to plan a set of experiments to investigate a complex problem in materials processing.

The **objectives** for Part 2 of this exercise are:

- Use a set of real experimental data
- Work out what factors are actually important
- Write a short group report

EXP2-4 What we will do



Part 2: (Week 14)

I will give you some real experimental data.
By making some simple calculations you will be able to tell me what actually changed the processing.

First, I will introduce the idea of design of experiments

You have seen from Part 1 of this exercise that it is **not so easy** to plan a complex set of experiments!

There are 'experimental designs' that can be used for different types of investigations

Design of experiments



- Compare a **control** (well-known) sample to a **variation** (e.g. used to test the efficacy of a new pharmaceutical)
- **Benchmark** – test each variable against a gold standard (like the first but with multiple variables)
- **One Factor At a Time (OFAT)**
change each variable keeping all others constant
(no gold standard – assumes all variables are independent)
- **Factorial**
test ***all*** the possible factors, ***all at the same time*** !
without making any assumptions

Design of experiments



Factorial designs are often used for process control

In process control problems (like the one in this exercise) there may be several factors that can influence the process

- These factors may have a large or a small effect on the product performance
- The factors may be independent or may interact with each other
- Some may be critical to the process and hence the product performance or critical to the product's reliability

Factorial designed experiments



Plackett and Burnham 1930's, Cambridge, UK

2 mathematicians developed factorial designed experiments

Factorial designed experiments:

- can test several factors simultaneously
- all experiments in the design **must** be completed (the statistical basis of the design relies on this)
- there is no 'normal' or 'control' sample
all of the factors are tested against all others
- **statistical methods** are used to examine the results

Simple example



2 factor – 2 level experiment

Choosing the factors and levels carefully is very important –
do not assume that anything can be ignored

Identify **factors** that *could* be important

- set 2 **levels** for each **factor** (these must be realistic and achievable)
e.g. for 2 *Factors* at 2 *Levels*

Factors	Level	
	+	-
A	X	Y
B	high	low

Experimental design



Measure a property that is representative of the product application

	Factors	
	A	B
Run 1	+	+
Run 2	-	+
Run 3	+	-
Run 4	-	-

Outcomes	
Y	
45	35
44	56
27	33
77	63

Outcomes are measures of a relevant response from the 'system'

Outcomes must be **free variables** (i.e. can have any value)

Repeat the test to estimate how **reliable** the '**outcome**' is (not the test)

Mean outcome and range



Calculate the mean outcome Y' and the range of values R

	Factors	
	A	B
Run 1	+	+
Run 2	-	+
Run 3	+	-
Run 4	-	-

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

Y' is the mean outcome

R is the range of Y (i.e. max-min)

R' is the mean range

Calculate 'main effects'



For each factor add up the mean outcome values for the cases:
when the factor was + and when the factor was -

	Factors	
	A	B
Run 1	+	+
Run 2	-	+
Run 3	+	-
Run 4	-	-
Y'	SUM +	70
	SUM -	120

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

Calculate 'main effects'



For each factor add up the mean outcome values for the cases:
when the factor was + and when the factor was -

	Factors	
	A	B
Run 1	+	+
Run 2	-	+
Run 3	+	-
Run 4	-	-
Y'	SUM +	70 90
	SUM -	120 100

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

Calculate interactions



Interactions between different factors can also be calculated

		Factors			Outcomes			
		A	B	AB	Y		Y'	R
Run 1		+	+	+	45	35	40	10
Run 2		-	+	-	44	56	50	12
Run 3		+	-	-	27	33	30	6
Run 4		-	-	+	77	63	70	14
Y'	SUM +	70	90	110	Total Y		190	/
	SUM -	120	100	80	Ave R'		/	10.5

+ and + = +
 + and - = -
 - and - = +

Identify effects and interactions



For each factor calculate the *Effect*
compare this to the *Random variation*

	Factors		
	A	B	AB
Run 1	+	+	+
Run 2	-	+	-
Run 3	+	-	-
Run 4	-	-	+
Y'	SUM +	70	90
	SUM -	120	100

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

Identify effects and interactions



For each factor calculate the *Effect*
compare this to the *Random variation*

	Factors		
	A	B	AB
Run 1	+	+	+
Run 2	-	+	-
Run 3	+	-	-
Run 4	-	-	+
Y'	SUM +	70	90
	SUM -	120	100

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

$N = \text{number of values in each sum (i.e. 2)}$

$\text{Random variation} = (R' \times 2)/N = 10.5$

$\text{Difference} = |(\text{Sum} +) - (\text{Sum} -)|$

Identify effects and interactions



For each factor calculate the *Effect*
compare this to the *Random variation*

	Factors		
	A	B	AB
Run 1	+	+	+
Run 2	-	+	-
Run 3	+	-	-
Run 4	-	-	+
Y'	SUM +	70	90
	SUM -	120	100
	Diff	50	10

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

$N = \text{number of values in each sum (i.e. 2)}$

Random variation $= (R' \times 2)/N = 10.5$

$\text{Difference} = |(\text{Sum} +) - (\text{Sum} -)|$

Effect $= |\text{Difference}|/N$

Identify effects and interactions



For each factor calculate the *Effect*
compare this to the *Random variation*

		Factors		
		A	B	AB
	Run 1	+	+	+
	Run 2	-	+	-
	Run 3	+	-	-
Run 4	-	-	+	
Y'	SUM +	70	90	110
	SUM -	120	100	80
	Diff	50	10	30
	Effect	25	5	15

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

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Random variation $= (R' \times 2)/N = 10.5$

Difference $= |(\text{Sum} +) - (\text{Sum} -)|$

Effect $= |\text{Difference}|/N$

Identify effects and interactions



For each factor calculate the *Effect*
compare this to the *Random variation*

	Factors		
	A	B	AB
Run 1	+	+	+
Run 2	-	+	-
Run 3	+	-	-
Run 4	-	-	+
Y'	SUM +	70	90
	SUM -	120	100
	Diff	50	10
	Effect	25	5

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

$N = \text{number of values in each sum (i.e. 2)}$

Random variation $= (R' \times 2)/N = 10.5$

$\text{Difference} = |(\text{Sum} +) - (\text{Sum} -)|$

Effect $= |\text{Difference}|/N$

Effect > Random variation, it is real.

Identify factors affecting reliability



The same calculations can be done for the range values.

This reveals which factors cause the process to become unreliable.

	Factors		
	A	B	AB
Run 1	+	+	+
Run 2	-	+	-
Run 3	+	-	-
Run 4	-	-	+

Y'	SUM +	70	90	110
	SUM -	120	100	80
	Diff	50	10	30
	Effect	25	5	15

R	SUM +	16	22	24
	SUM -	26	20	18
	Diff	10	2	6
	Effect	5	1	3

Outcomes			
Y		Y'	R
45	35	40	10
44	56	50	12
27	33	30	6
77	63	70	14
Total Y		190	/
Ave R'		/	10.5

$N = \text{number of values in each sum (i.e. 2)}$

Random variation $= (R' \times 2)/N = 10.5$

Difference $= |(\text{Sum} +) - (\text{Sum} -)|$

Effect $= |\text{Difference}|/N$

Effect > Random variation, it is real.

Summary



- Factorial designed experiments are used when many different factors may be involved
- All factors are investigated without prejudice
- The factors effecting the outcome of a process are identified including interactions between factors
- Also the factors causing the process to be unreliable can be identified from analysis of the range of values
- These methods are used in '**Statistical process control**' to control manufacturing processes and ensure quality products
- They can identify the critical process steps that need to be tightly controlled to make the process (and so the product) reliable

EXP2-4 What you will do



Read the instructions for Part 2!

Use the **data from the factorial experiment**

Calculate the average outcome and range

Find the **effects and interactions**

Write a group report:

- Tell me what you find?
- Compare this to the 'Blamestorming Conversation'
- Would your Group's experimental plan have found this?
- What should be done next?

EXP2-4 Schedule



P Groups

Date:	Time:	Place:	Activity:
Friday 16 th	10:30-12:10	A310	Introduction to Part 2 of EXP2-4
Monday 19 th	08:30-10:10	A310	Data processing & analysis
Wednesday 21 st	08:30-10:10		Blamestorming
Friday 23 rd	10:30-12:10		Complete and Submit Report

M Groups

Date:	Time:	Place:	Activity:
Friday 16 th	08:30-10:10	A310	Introduction to Part 2 of EXP2-4
Monday 19 th	10:30-12:10	A310	Data processing & analysis
Wednesday 21 st	10:30-12:10		Blamestorming
Friday 23 rd	08:30-10:10		Complete and Submit Report